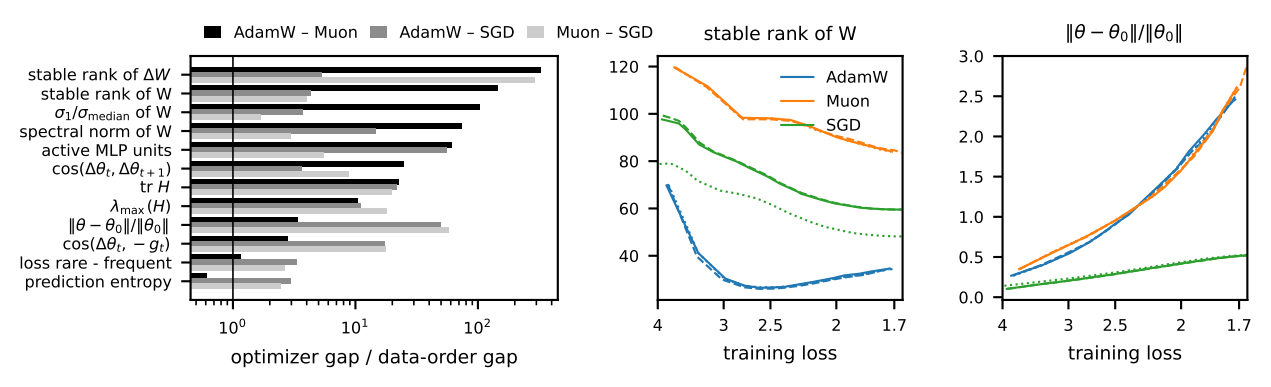
AdamW – Muon
AdamW – SGD
Muon – SGD
stable rank of ΔW
stable rank of W
$\sigma_1/\sigma_{\text{median}}$ of W
spectral norm of W
active MLP units
$\cos(\Delta\theta_t, \Delta\theta_{t+1})$
tr H
$\lambda_{\max}(H)$
$\|\theta-\theta_0\|/\|\theta_0\|$
$\cos(\Delta\theta_t, -g_t)$
loss rare - frequent
prediction entropy
10^0
10^1
10^2
optimizer gap / data-order gap
stable rank of W
AdamW
Muon
SGD
120
100
80
60
40
4
3
2.5
2
1.7
training loss
$\|\theta-\theta_0\|/\|\theta_0\|$
3.0
2.5
2.0
1.5
1.0
0.5
0.0
4
3
2.5
2
1.7
training loss